

TRIO PRAWIRO NEGORO

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SUMMARY

Highly skilled Hardware & IoT Engineer with a 3.73 GPA and extensive experience in the end-to-end development of industrial-grade electronic systems. Expert in designing DFM-compliant PCB layouts (SMD/SMT) and 3D mechanical mockups using SketchUp to optimize hardware-software integration. Proven track record in developing complex IoT ecosystems, including RFID UHF asset tracking, low-power access control systems, and ESP32-based automation. Proficient in implementing advanced control methods like Fuzzy Logic and long-range communication via LoRaWAN. A certified CCNA Network Engineer with strong technical leadership skills, dedicated to bridging physical hardware with digital networks through robust research, middleware engineering, and API integration

EXPERTISE

- Fullstack Developer web
- Microsoft Office
- Android Studio
- PHP
- Java
- Arduino IDE
- CSS
- Operating System
- Corel Draw
- AutoCAD
- Sketchup
- Fiber Optic
- Network Installation
- C/C++

WORK EXPERIENCE

PT Data Nusantara Adikarya (Caliana)
Hardware IoT Engineer **Dec 2025 - Present**

- API Research & Web Integration: Conducted extensive research on Access Control API protocols to enable seamless integration with internal web platforms. Developed open-source API documentation for third-party compatibility and performed rigorous endpoint testing using Postman to ensure robust device-to-web communication.
- Industrial Flowmeter System: Engineered a high-precision water tank calculation system by selecting and integrating industrial Flowmeter sensors with PLC controllers. Configured system logic based on client specifications and executed end-to-end trial phases to ensure measurement accuracy.
- Middleware Development & Networking: Developed a custom middleware system to bridge Access Control devices with public networks. Configured hardware services, including LED indicators and button triggers, and designed custom PCB plating enclosures to house the middleware components.
- 3D Modeling & Mechanical Design: Developed high-fidelity 3D Mockups for Mini-Kiosk units using SketchUp, specifically optimizing the spatial placement of OCR cameras for maximum scan accuracy. Refined the mechanical design to include ergonomic component housing, thermal ventilation, and cable management paths.

PT Clavata Extra Sukses (ID-GROW)
Hardware IoT Engineer **Nov 2024 - Dec 2025**

- RFID UHF Ecosystem: Designed and deployed end-to-end asset tracking and inventory systems using RFID UHF (Handheld & Fixed)
- Advanced PCB Design: Developed high-density SMD/SMT PCB layouts following DFM standards to ensure production efficiency.
- Low-Power IoT: Engineered an ESP32-based access control and automated laundry system, optimizing for Low Quiescent Current to maximize battery life.
- Product Prototyping: Managed full product lifecycles, including casing mock-ups, antenna range testing, and hardware-software integration.
- Technical Leadership: Lead the hardware team in project timelines and interdepartmental coordination to ensure 100% successful system deployment.
- Field Maintenance: Performed routine troubleshooting and preventive maintenance on client hardware to minimize system downtime.

Solution Project (Freelancer) **Jan 2024 - Present**

- Creating a smart PLTS web-based tool and monitoring system using the fuzzy logic method.
- Creating a system for hatching chicken eggs and creating controlling and monitoring applications using the fuzzy method.
- Create Android tools and applications to monitor accident-prone areas using tilt sensors and GPS as well as sound using geofencing technology method.
- Creating tools and web for monitoring water cubication of PDAM meters using LoRaWAN technology and OCR.
- Create tools and websites for monitoring and controlling automatic watering of orchid plants using LoRa technology and fuzzy logic method.
- Create an IoT-based canopy tool using voice recognition, rain, temperature and LDR sensors based on an Android application
- Create a system of tools and Android applications to control temperature and PH in glofish ponds using the fuzzy logic method
- Creating a smart basket system tool to optimize queues and transaction efficiency in supermarkets using web-based barcode and RFID scanning sensors

PT Icon Plus PLN (ICONNET) Surabaya
Field Support (intern) **July 2023 - Dec 2023**

- Network Infrastructure Quality Control: Performed comprehensive quality control for network device installations, including POP, FDT, and FAT verification to ensure compliance with technical standards.
- Network Deployment & Configuration: Assisted in the physical installation and configuration of Fortinet SD-WAN routers and performed post-installation network connectivity tests.
- Technical Asset Management: Managed hardware dismantling procedures and maintained OSP (Outside Plant) accessories such as deaden suspends for network durability.

PT Hendevane Indonesia (Jakarta, Indonesia)
Fullstack Web Developer (intern) **February 2023 - June 2023**

- Web Application Development: Developed responsive web applications using Laravel, PHP (RESTful API), and JavaScript, while managing databases with MySQL/MariaDB.
- Frontend & UI/UX: Implemented modern UI/UX designs using HTML, CSS, Bootstrap, and multimedia integration.
- School Management System: Designed and deployed a comprehensive website for Baiturrahim Islamic Boarding School, featuring dedicated admin and registrant dashboards.
- Version Control: Utilized GitHub for version control and collaborative development

ORGANIZATIONAL EXPERIENCE

Electrical Student Association (Malang, Indonesia) **January 2020 - August 2021**
Orginizing Commite

- Served as Chief Committee for the "Kuliah Tamu" event, managing event logistics and coordination for the Electrical Engineering department.
- Leading routine religious and community-building programs for association members.
- Actively contributed to planning and executing major departmental events and initiatives.

Workshop Electro Malang State Polytechnic (Malang, Indonesia) **March 2021 - March 2022**
General Division

- organize workshop activities regarding competitions and activities within the WSEC environment

EDUCATION

D-IV Electrical Engineering - Digital Telecommunication Network
Malang State Polytechnic **2020 - 2024**

- Participated and active in the Electro Workshop organization.
- Participated in the national Scientific Writing Competition.
- Participated and active in the Electrical Student Association.

Sumenep State Aliyah Madrasah

- 3rd place in the district level AKSIOMA badminton competition **2017 - 2020**

ADDITIONAL INFORMATION

Awards - Malang State Polytechnic

- Achievement of the best Achievement Index for semesters 2,3,4,5. **July 2020 - 2024**
- 1st place in PKM (Student Creativity Program) majoring in Electrical Engineering.
- passed the National Ministry of Education and Culture's PKM funding.
- 3rd place winner of the LKTI Faperta Expo II competition

Language

- English
- Indonesia

Softskills

- Time Management
- TeamWork
- Problem Solving
- Communication

Certifications & Training

- CCNA Network Engineer: Trained in Switching, Routing, Wireless Essentials, Enterprise Networking, Security, and Automation.
- Access Network Planner: Certified by BNSP (2024).
- Fullstack Web Developer: Competency Certificate (2023).
- Basic PLC: Technical Training (2024).